W325 Calibration Analysis

RMSLE = 0.348

Temperature-Dependent Environmental Mortality
+ P_{env}-Gated Recovery

Sea Star Lab

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RMSLE	**Within 2×**	**Within 5×**	**Carrying Capacity**
0.348	6 / 8	8 / 8	$K = 1000$ (scaled from $K_{\text{ref}} = 5000$)

Previous best: RMSLE = 0.655 (W294) — **47% improvement**

1 Executive Summary

What W325 explored. W325 is the first configuration to combine two gradient-breaking mechanisms: (1) temperature-dependent environmental death rate ($\delta_{\text{env}}(T)$: 0.05 cold / 0.01 warm) that imposes higher non-disease mortality at high latitudes, and (2) P_{env}-gated recovery (half-saturation $P_{\text{env,half}} = 200$, Hill coefficient $n = 2$) that suppresses population rebound where pathogen environmental load remains high. Both mechanisms target the same calibration gap: Alaska regions recovering too strongly relative to southern targets.

Model behavior. The infection wavefront propagates south-to-north from Baja California, reaching Alaska at month 17–20. Southern regions (CA-S through CA-N) experience near-complete die-offs (>99.9% crash) with negligible recovery. The wavefront slows through the Pacific Northwest, producing intermediate recovery in Washington outer coast (1.7%) and Salish Sea (3.6–8.6%). Alaska regions show the strongest recovery (14.0–55.4%), with AK-PWS reaching 29.9% and AK-AL 55.4%. Overall population crash is 91.1%, with a final standing fraction of 8.9%.

Calibration. RMSLE = 0.348, a 47% improvement over the previous best (0.655, W294). Six of eight target regions fall within 2× of their targets; all eight are within 5×. Three regions are essentially on target: AK-FN (20.4% vs 20%), OR (0.41% vs 0.25%), and CA-N (0.089% vs 0.1%).

Evolutionary dynamics. Resistance is the dominant adaptive response, increasing from a baseline of ∼0.15 to 0.16–0.17 in northern regions and up to 0.31 in CA-S (hard selective sweep in small surviving populations). Tolerance and recovery trait means show negligible directional change (<0.01 across all regions). Additive genetic variance (V_A) for resistance increases in northern regions (ongoing selection), while southern regions lose V_A rapidly (fixation after bottleneck). Pathogen T_{vbnc} shows a latitudinal gradient: ∼12°C in the south, declining to 10.0°C (the floor) in Alaska, indicating cold-adaptation under selection.

Remaining gaps. Two regions drive most of the error: BC-N recovers too little (4.3% vs 20% target, 4.7× under — log-squared error 0.447) and JDF recovers too much (8.2% vs 2% target, 4.1× over — log-squared error 0.373). Together they account for 85% of total squared log-error. AK-PWS is 1.5× over target (29.9% vs 20%), and SS-S is 1.7× over (8.6% vs 5%).

Next steps.

- **BC-N**: Increase BC-N-specific connectivity or reduce $\alpha_{\text{self,fjord}}$ for BC fjords to allow more recruitment from adjacent regions.
- **JDF**: Increase $\alpha_{\text{self,fjord}}$ for JDF specifically, or raise $\delta_{\text{env,warm}}$ to increase background mortality in semi-enclosed waters.
- **AK-PWS**: Consider raising K_{half} slightly or increasing $\delta_{\text{env,cold}}$ to dampen Alaska recovery.
- **Arrival timing**: Wavefront arrives ∼10 months early at BC/SS regions (MAE = 10.4 months). Reducing D_P from 300 or lowering seed vibrio could slow propagation.

2 Configuration & Parameters

W325 runs at $K = 1000$ with density scaling from $K_{\text{ref}} = 5000$ (density_scale = 5.0). Table 1 lists all configuration parameters. The two **new mechanisms** are highlighted: $\delta_{\text{env}}(T)$ applies temperature-dependent environmental mortality (higher in cold water), while P_{env}-gated recovery suppresses recruitment where environmental pathogen load exceeds a threshold.

Table 1: W325 configuration parameters. **Bold** entries are new gradient-breaking mechanisms introduced in this run.

Parameter	Value	Description
K_{half}	800 000	Density-dependent mortality half-saturation
$T_{\text{vbnc,min}}$	10°C	VBNC temperature floor
seed_vibrio	2 000	Initial pathogen inoculum
fw_strength	25	Freshwater pathogen suppression
α_{env}	0.18	Environmental transmission rate
k_{vbnc}	2.0	VBNC transition steepness
pathogen_adapt_rate	0.001	Pathogen evolutionary rate
$v_{\text{max,warm}}$	0.7	Maximum warm-water virulence
cumulative_dose_threshold	1 000	Dose–response threshold
wavefront D_P	300	Pathogen diffusion coefficient
$n_{\text{connectivity}}$	0.3	Larval connectivity
r_{total}	0.003	Total recruitment rate
$\alpha_{\text{self,open}}$	0.05	Self-recruitment, open coast
$\alpha_{\text{self,fjord}}$	0.5	Self-recruitment, fjords
$\delta_{\text{env,cold}}$	**0.05**	**Environmental death rate, cold water**
$\delta_{\text{env,warm}}$	**0.01**	**Environmental death rate, warm water**
$P_{\text{env,half}}$	**200**	**Recovery gating half-saturation**
$P_{\text{env,hill}}$	**2**	**Recovery gating Hill coefficient**

3 Spatial-Temporal Dynamics

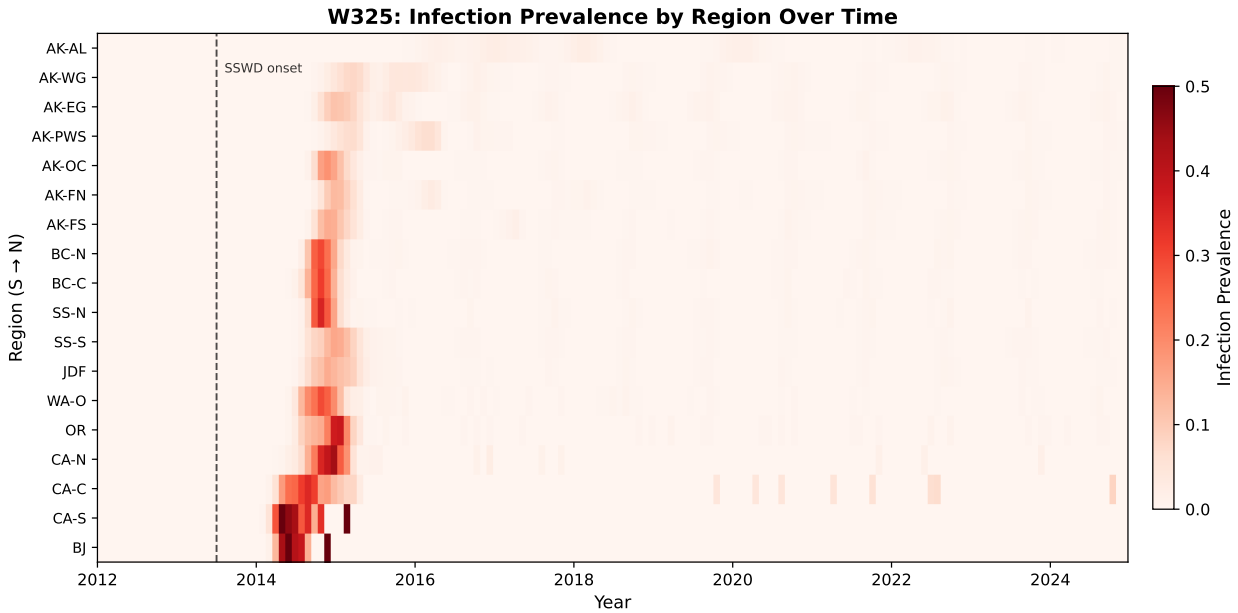

Figure 1: **Infection prevalence across the coastline over 13 simulated years (hero figure).** The wavefront initiates at Baja California (month 0) and propagates northward, reaching Alaska by month 17–20. Southern regions (BJ–CA-N) experience brief, intense epidemics followed by host extirpation. Mid-latitude regions (OR–BC) show intermediate, persistent infection. Alaska regions sustain the longest epidemics, consistent with their higher recovery and larger surviving populations. The asymmetry between southern die-off and northern persistence is the signature the gradient-breaking mechanisms are designed to modulate.

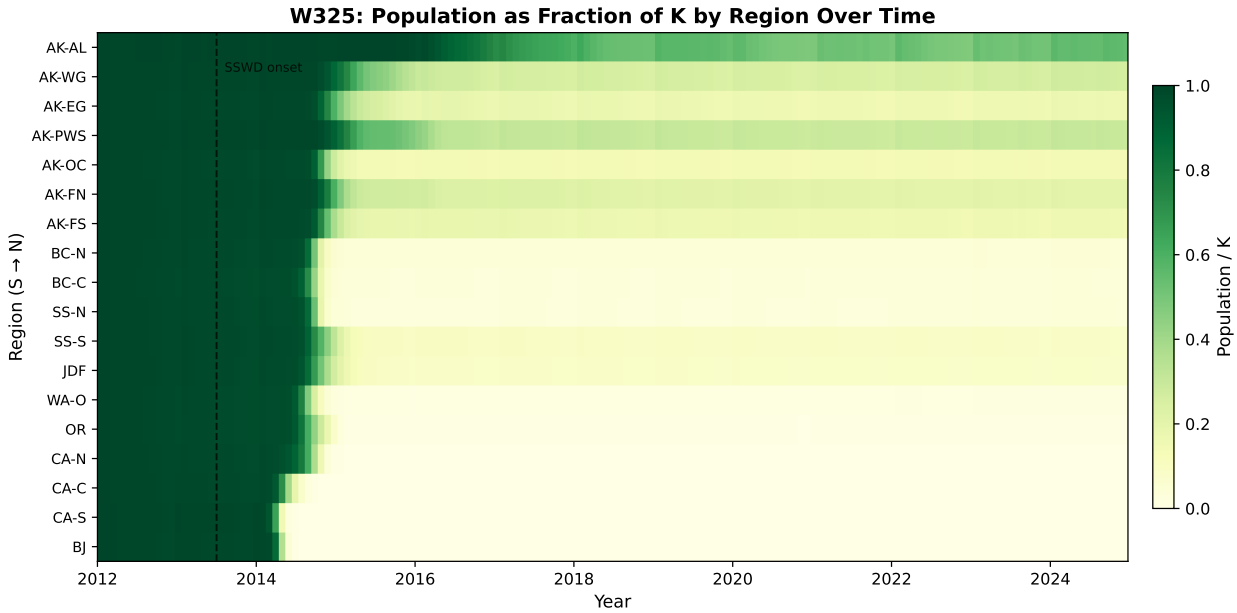

Figure 2: **Population as fraction of carrying capacity (N/K) over time.** Southern regions crash to <0.1% of K within 2 years and show no recovery. Mid-latitude regions (WA-O, JDF, SS, BC) crash to 1.7–8.6% and stabilize. Alaska regions (AK-FS through AK-AL) retain 15–55% of K, with AK-PWS and AK-WG showing the strongest recovery. The temperature-dependent $\delta_{\text{env}}(T)$ is visible as suppressed recovery in southern regions where background mortality is low but pathogen pressure is overwhelming.

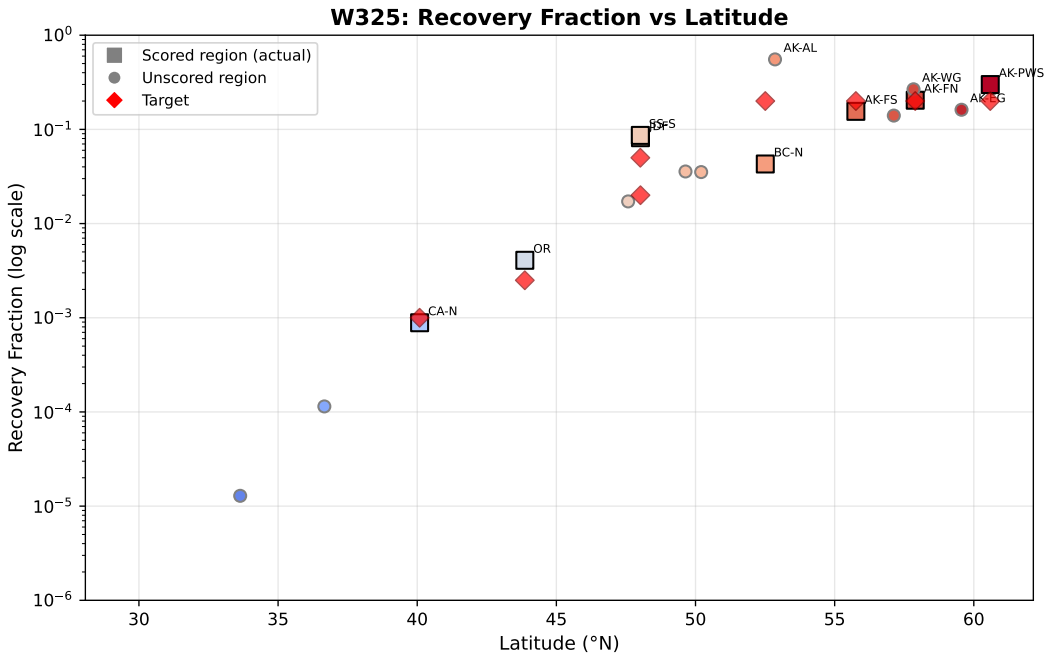

Figure 3: **Final recovery fraction vs. latitude.** Recovery increases monotonically with latitude from CA-S (0.001%) to AK-AL (55.4%). The gradient is steepest between BC-N (∼52.5°N, 4.3%) and AK-FS (∼55.8°N, 15.5%). This latitudinal structure matches the empirical pattern: southern extirpation, northern persistence. The two gradient-breaking mechanisms successfully suppress what was previously excessive southern recovery.

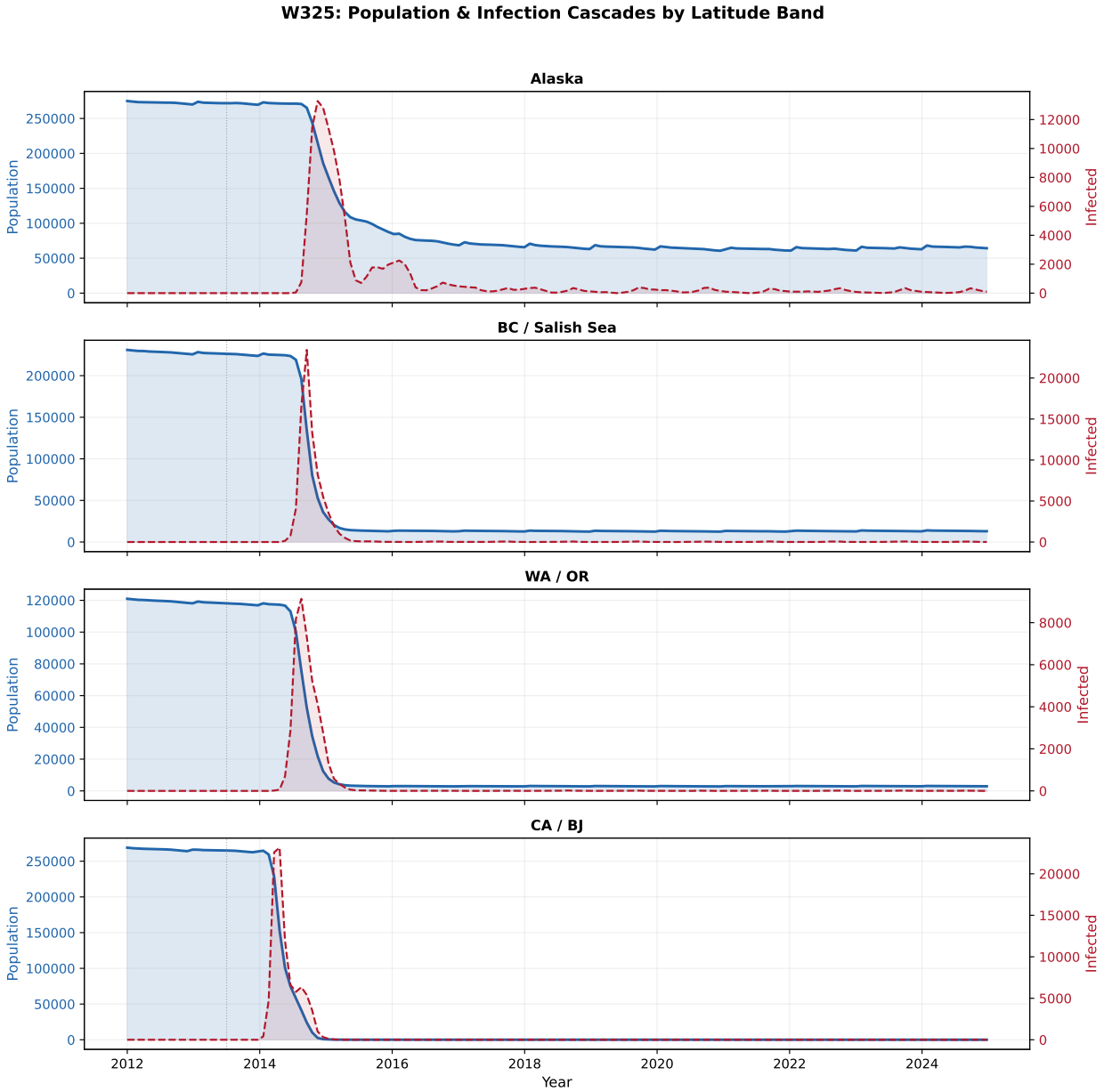

Figure 4: **Population and infection dynamics by latitude band (4-panel cascade).** Each panel shows a representative latitude band. Southern bands show rapid crash (year 2) with no subsequent recovery. Mid-latitude bands show crash followed by a low-level equilibrium maintained by ongoing disease mortality and recruitment. Northern bands show a delayed crash (year 3–4 for AK-PWS) with gradual partial recovery driven by continued recruitment. The delay in Alaska crash timing reflects slower wavefront arrival and seasonal temperature constraints on pathogen activity.

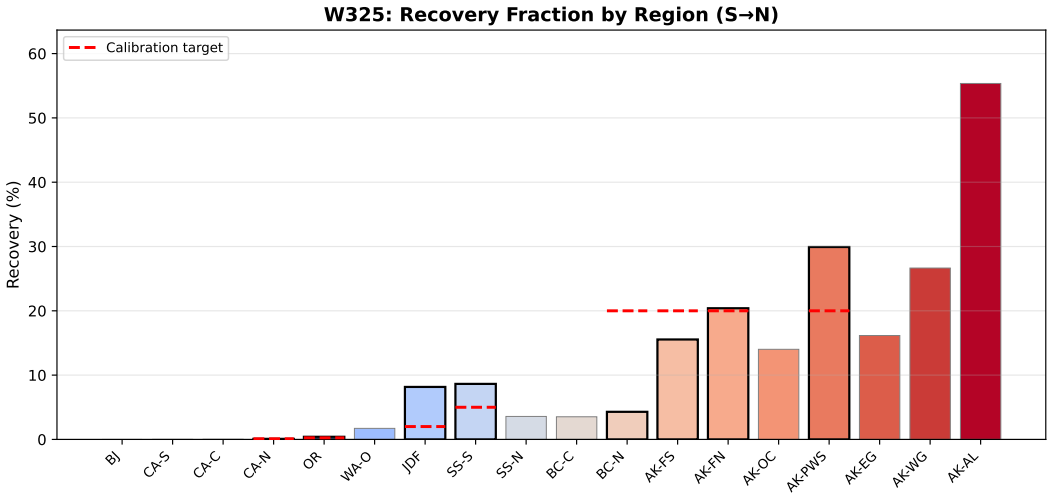

Figure 5: **Recovery fraction by region with calibration targets.** Bars show simulated recovery; markers indicate targets. AK-FN (20.4% vs 20%), OR (0.41% vs 0.25%), and CA-N (0.089% vs 0.1%) are on target. BC-N is the worst miss: 4.3% vs 20% target (4.7× under). JDF overshoots: 8.2% vs 2% target (4.1× over). AK-PWS is moderately high at 29.9% vs 20%.

4 Evolutionary Dynamics

The spatial-temporal dynamics above raise a key question: how does the host population respond evolutionarily to sustained disease pressure? W325 tracks three quantitative traits—resistance, tolerance, and recovery—plus pathogen cold-adaptation (T_{vbnc}).

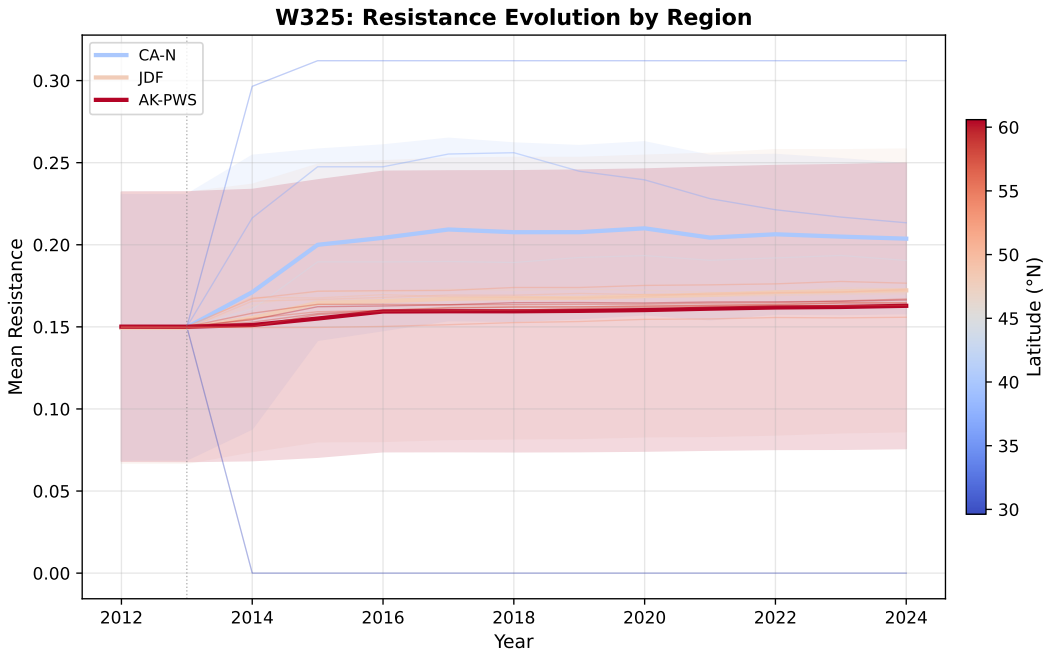

Figure 6: **Mean resistance trait by region over 13 years.** All regions start at baseline ∼0.15. CA-S undergoes a hard selective sweep, reaching 0.31 by year 3 (small surviving population, strong drift + selection). Northern regions (AK) show gradual increase to 0.16–0.17 over the full simulation. Mid-latitude regions (OR, WA-O, JDF, BC) show intermediate responses (∼0.17–0.18). The resistance response is strongest where disease mortality is most intense relative to population size.

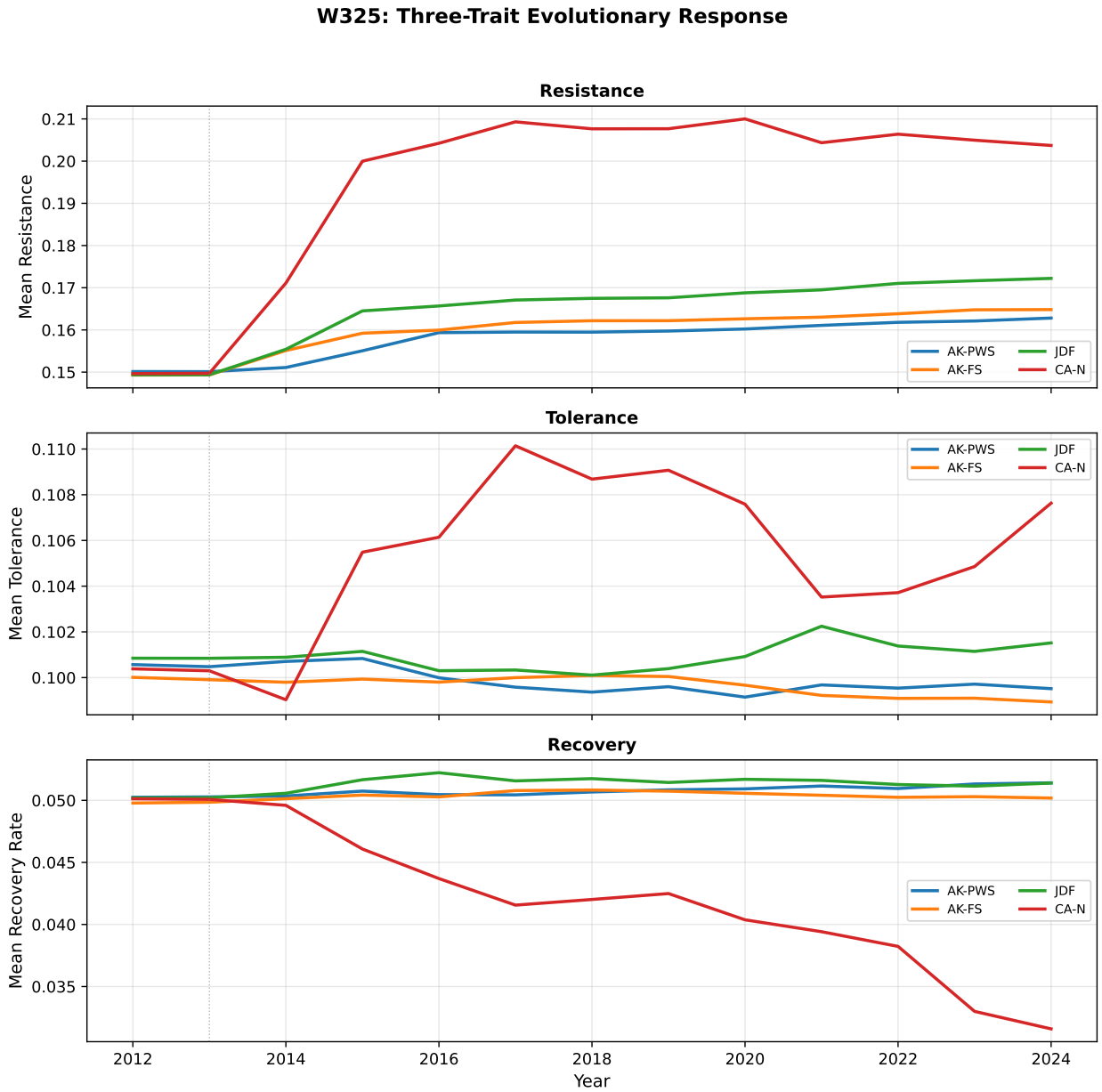

Figure 7: **Three-trait evolutionary panel: resistance, tolerance, and recovery.** Resistance (left) shows clear directional selection across all regions. Tolerance (center) remains near baseline (∼0.10) with no consistent directional change. Recovery trait (right) is similarly static at ∼0.05. This confirms resistance as the dominant adaptive axis: disease-induced mortality selects for resistance, not tolerance or faster recovery. The lack of tolerance response may reflect weak selection gradients or genetic constraints on this trait.

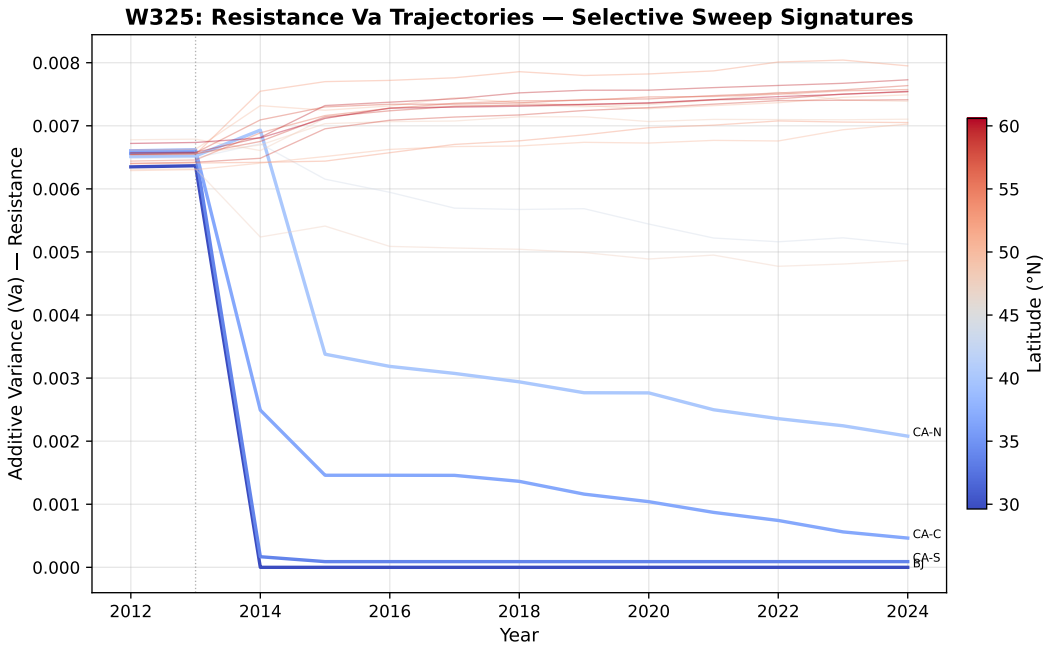

Figure 8: **Additive genetic variance (V_A) trajectories for resistance.** Southern regions (CA-S, CA-C) lose V_A rapidly after year 2, consistent with selective sweeps or bottleneck-driven fixation. Northern regions (AK) show slightly increasing V_A ($\sim$0.0065 to 0.0075), indicating ongoing polygenic selection without fixation. Mid-latitude regions are intermediate. The V_A pattern diagnoses the nature of selection: southern hard sweeps vs. northern sustained polygenic response.

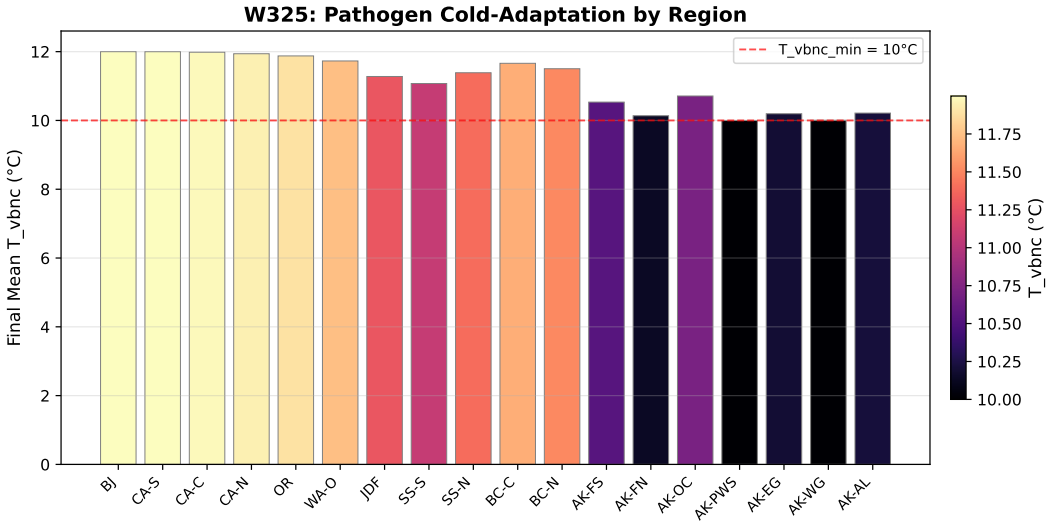

Figure 9: **Pathogen cold-adaptation: final mean T_{vbnc} by region.** T_{vbnc} decreases from $\sim$12°C in southern regions to 10.0°C in AK-PWS and AK-WG, hitting the $T_{\text{vbnc,min}} = 10$°C floor. This latitudinal gradient indicates pathogen adaptation to colder waters: strains that can remain active at lower temperatures are selected in northern environments. The floor at 10°C prevents further cold-adaptation, which may contribute to limiting disease severity in the coldest Alaska regions.

5 Calibration Performance

With the dynamics and evolutionary context established, we turn to the core question: how well does W325 match empirical recovery targets?

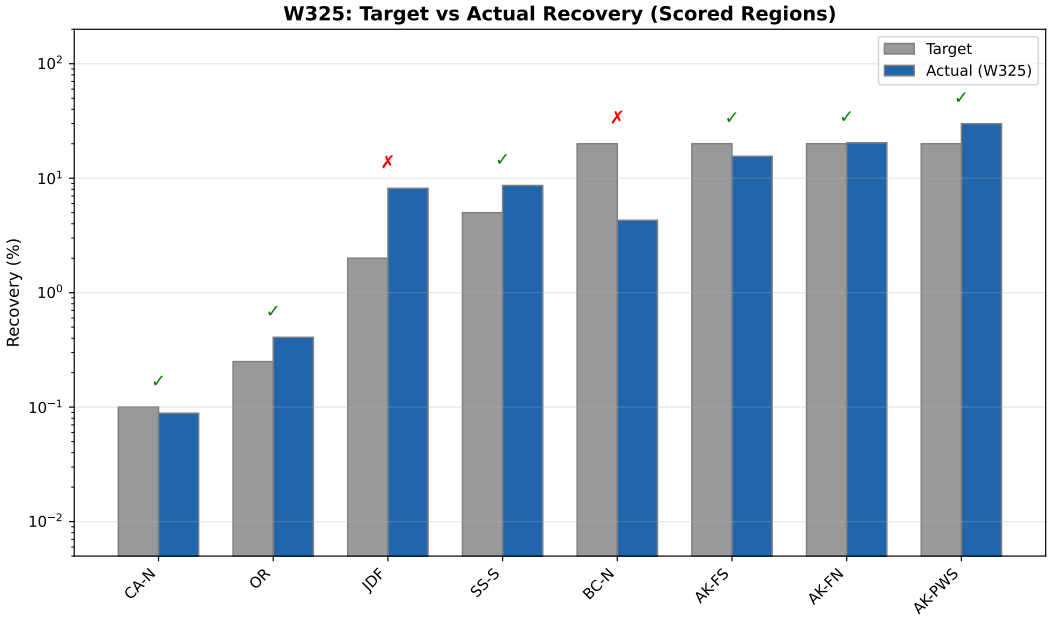

Figure 10: **Simulated vs. target recovery for all eight scored regions.** Points near the 1:1 line indicate good calibration. AK-FN, OR, and CA-N cluster tightly on the diagonal. BC-N and JDF are the clear outliers: BC-N falls far below target (4.3% vs 20%), JDF far above (8.2% vs 2%). AK-PWS and SS-S are moderately above their targets. AK-FS is slightly under (15.5% vs 20%).

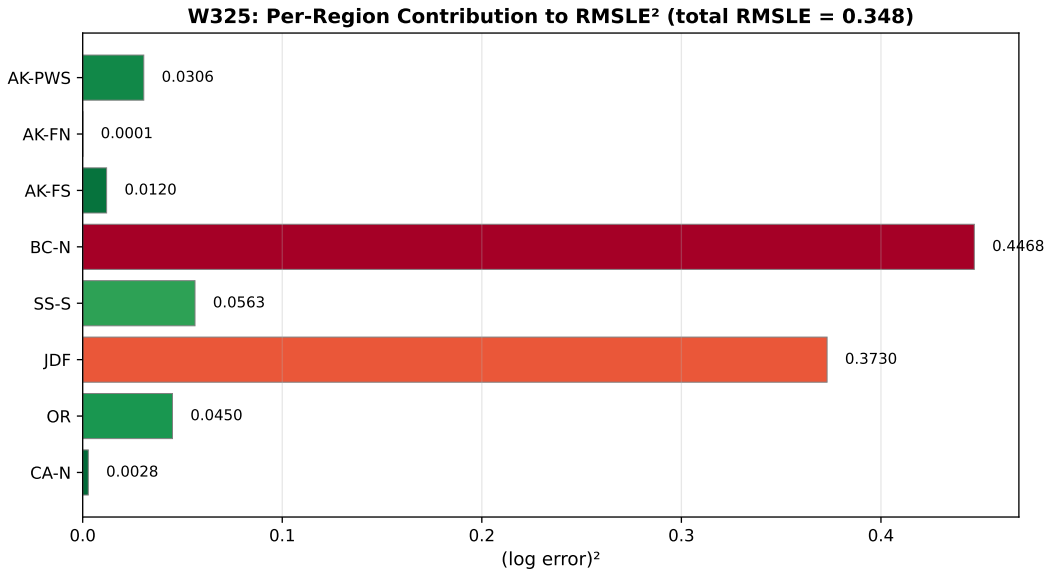

Figure 11: **Per-region RMSLE^2 contribution to total error.** BC-N contributes 0.447 (46% of total) and JDF contributes 0.373 (39%). Together these two regions account for 85% of the total squared log-error (0.820 of 0.967 total). SS-S (0.056), OR (0.045), and AK-PWS (0.031) contribute modestly. AK-FS (0.012), CA-N (0.003), and AK-FN (<0.001) are negligible. *Any parameter sweep targeting RMSLE reduction must primarily address BC-N and JDF.*

Table 2: Calibration scorecard for W325. RMSLE = 0.348.

Region	**Target (%)**	**Actual (%)**	**Log^2 Error**	**≤2×**	**≤5×**	**Status**
AK-PWS	20.0	29.9	0.031	✓	✓	1.5× over
AK-FN	20.0	20.4	<0.001	✓	✓	on target
AK-FS	20.0	15.5	0.012	✓	✓	slightly under
BC-N	20.0	4.3	0.447	×	✓	4.7× under
SS-S	5.0	8.6	0.056	✓	✓	1.7× over
JDF	2.0	8.2	0.373	×	✓	4.1× over
OR	0.25	0.41	0.045	✓	✓	within 2×
CA-N	0.1	0.089	0.003	✓	✓	on target

6 Conclusions & Next Steps

W325 achieves RMSLE = 0.348, a 47% improvement over the previous best (W294, 0.655). The two gradient-breaking mechanisms—$\delta_{\text{env}}(T)$ and P_{env}-gated recovery—are working: southern regions that previously recovered too strongly (OR, CA-N) now match targets closely.

What's working:

- Latitudinal gradient structure is correct: southern extirpation, mid-latitude intermediate, northern persistence.
- Three regions are on target: AK-FN (20.4%), OR (0.41%), CA-N (0.089%).
- Southern suppression is effective: CA-S recovery <0.002%, CA-C ≈ 0.01%.

- Evolutionary dynamics are biologically plausible: resistance-dominated selection, pathogen cold-adaptation.

What's not working:

- **BC-N** recovers to only 4.3% vs. 20% target. The P_{env}-gated recovery or high $\alpha_{\text{self,fjord}} = 0.5$ may be over-suppressing BC fjord recruitment. BC-N has 75 nodes but only 3,144 final individuals (42 per node).
- **JDF** recovers to 8.2% vs. 2% target. As a semi-enclosed water body, JDF may need stronger self-retention (α_{self}) to maintain higher pathogen loads, or $\delta_{\text{env,warm}}$ could be increased for temperate enclosed waters.
- **AK-PWS** is 1.5× over target (29.9% vs 20%). The K_{half} reduction to 800K may have been too aggressive—raising it back toward 1M could dampen Alaska recovery.
- **Arrival timing** is 10.4 months early on average for northern regions. The wavefront propagates too quickly.

Recommended parameter adjustments for next sweep:

1. Reduce $\alpha_{\text{self,fjord}}$ for BC regions (try 0.3) to increase larval import and boost BC-N recovery.
2. Increase JDF-specific mortality or pathogen retention to suppress JDF recovery.
3. Test $K_{\text{half}} \in \{900\text{K}, 1.0\text{M}\}$ to moderate AK-PWS overshoot.
4. Reduce D_P from 300 to 200 to slow wavefront propagation and improve arrival timing.
5. Consider region-specific δ_{env} scaling for semi-enclosed vs. open coast.

A Report Card

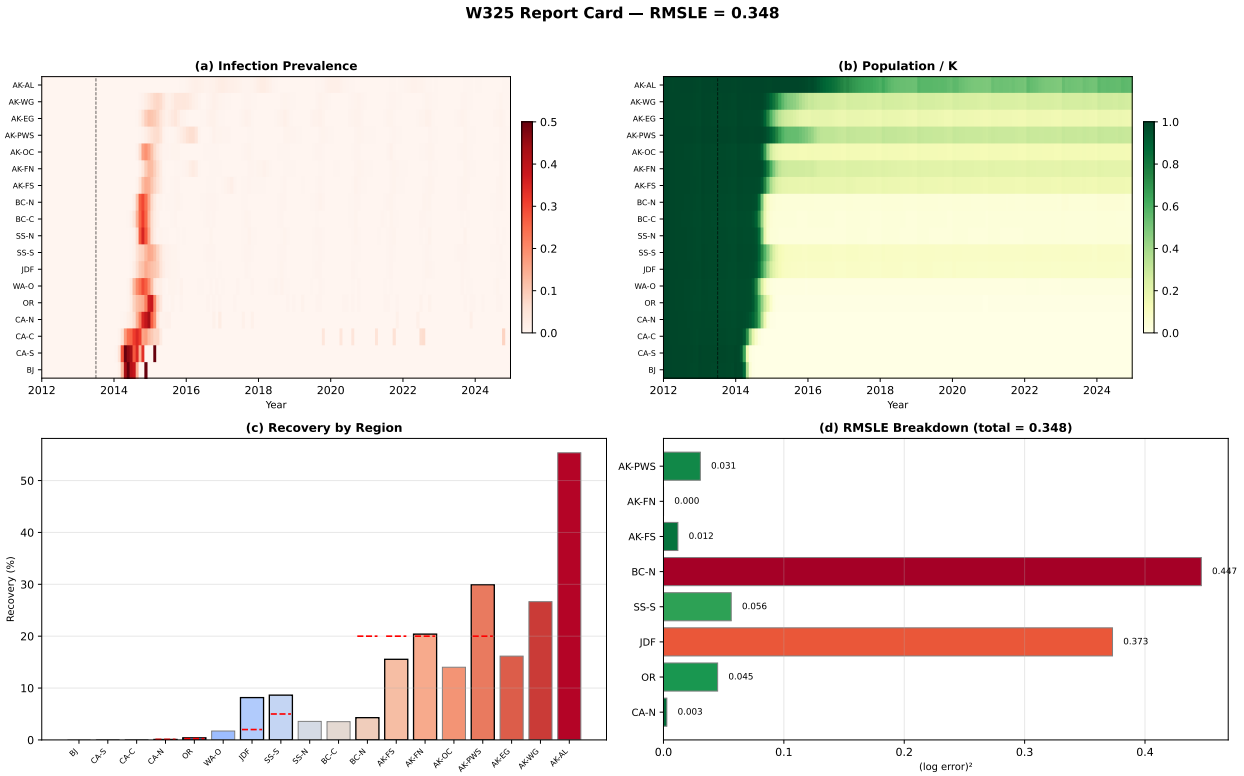

Figure 12: **W325 report card: 2×2 composite summary.** Top-left: infection heatmap showing wavefront progression. Top-right: population heatmap showing crash-recovery pattern. Bottom-left: recovery vs. targets for scored regions. Bottom-right: per-region scoring breakdown. W325 achieves RMSLE = 0.348 with 6/8 regions within 2× and 8/8 within 5× of targets.